Elana Simon

Email Google Scholar LinkedIn

Education

STANFORD UNIVERSITY PhD in Biomedical Informatics, 2022 - present
HARVARD UNIVERSITY B.A. in Computer Science, 2014 - 2018

Publications

Honeyman JN*, **Simon EP***, et.al. Science: 2014 Feb 28; 343(6174) doi: 10.1126/science.1249484. Detection of a Recurrent DNAJB1-PRKACA Chimeric Transcript in Fibrolamellar Hepatocellular Carcinoma. *co-first authors

Simon EP, et.al. Proceedings of the National Academy of Sciences: 2015 Nov 3; 112(44) doi: 10.1073/pnas.1424894112. Transcriptomic Characterization of Fibrolamellar Hepatocellular Carcinoma.

Shin JE, Riesselman AJ, Kollasch AW, McMahon C, **Simon EP**, et.al. Nature Communications: 2021 April 23; 12(2403) doi: 10.1038/s41467-021-22732-w. Protein design and variant prediction using autoregressive generative models.

Spyvee M, Kallenbach JM, Gupta A, **Simon EP**, Grand G. International Patent: Naptho [2,1 -D] Thiazole Derivatives, Compositions thereof and Methods of Treating Disorders.PCT:PCT/US2020/061796. Issued May 27, 2021

Farber BA, Lalazar G, **Simon EP**, et.al. Oncotarget. 2017 Dec 15;9(12):10211-10227. doi: 10.18632/oncotarget.23325. Non coding RNA analysis in fibrolamellar hepatocellular carcinoma.

Ahmad W*, **Simon EP***, Chithrananda S, Rhamsundar B. ELLIS Machine Learning for Molecule Discovery Workshop: 2021 Dec 13. ChemBERTa-2: Towards Chemical Foundation models, *co-first authors

Work Experience

Reverie Labs Senior Machine Learning Engineer, 2018 - 2022

- Implemented graph convolutional neural networks (GCNNs) and 3D CNNs to model small molecule ADMET properties and protein-ligand binding to design kinase inhibitors
- Used models to prioritize compounds for virtual screens and lead-optimization. Designed and used tools for de-novo molecular generation. Led to patented CDK9 inhibitor
- Optimized and analyzed methods for molecular pre-training with transformers leading to state of the art performance on multiple MoleculeNet benchmarks. Contributed pre-trained models to HuggingFace and gave talk at DeepChem meet-up
- Designed and implemented algorithm for determining ligand's binding mode in protein
- Wrote blog post using UMAP to analyze chemical datasets and detect biases. Performed detailed analysis with dimensionality reduction methods revealing unique properties of molecular embeddings

Harvard Medical School Student Researcher, Marks Lab, 2017

- Implemented autoregressive language models to model proteins
- Predicted phenotypic effects of genetic variation and generated novel protein sequences

Harvard College Computer Science Teaching Fellow, 2016-2018

- CS181 Machine Learning (2018), CSCI P-14200 Intro to Programming (2017), CS51 Abstraction/Design in Computation (2016)

Google Research Google Accelerated Science Intern, 2016

- Predicted bioactivity of molecules with GCNNs
- Discovered biases in experimental assays by analyzing model failure modes

Rockefeller University / Memorial Sloan Kettering Cancer Center Student Researcher, 2012-2016

- Initiated research project to characterize a rare liver cancer and analyze tumor transcriptome
- Discovered new oncogene (DNAJB1-PRKACA chimera) and inspired repurposing of therapeutics / an FDA approved clinical trial

Facebook iOS Development Summer Intern, FBU for Engineering, 2015

Awards / Honors

Gave keynote speech at American Association for Cancer Research in front of 15k cancer researchers

Won Inaugural AACR Young Champion in Cancer Research award, served as National Cancer Research Month Spokesperson

Introduced President Obama at White House Precision Medicine Announcement

Introduced the "Moonshot on Cancer" talks at the Vatican with Joe Biden and Pope Francis

Gave keynote speech at "Innovators in Adolescent and Young Adult Cancer" Case Comprehensive Cancer Center